

IMMUNODEFICIENCY

Primary ID: deficiency is the cause of the disease.

Secondary ID: Def. is a result of other diseases like infection.

Primary ID

B cell deficiency	50%
T cell deficiency	10%
Both B and T deficiency	20%
Phagocytic deficiency	18%
Complement deficiency	2%

All patients with recurrent infections should be screened for immunodeficiency

*recall: stem cells → pro B cells → B immature → B mature → plasma → memory
 *any deficiency in any enzyme in any part of B cells life would cause an immune deficiency in the whole B cell at DNA level

Lecture outline

- B cell ID
- Bruttons agammaglobulinemia
- Selective ID
- IgA selective ID
- DiGeorge syndrome
- T cell ID

IMMUNODEFICIENCY

Primary ID: deficiency is the cause of the disease.

Secondary ID: Def. is a result of other diseases like infection. ex viruses

* the most famous immune deficiency (secondary) is HIV (AIDS)

Primary ID

B cell deficiency 50%

T cell deficiency 10%

Both B and T deficiency 20%

Phagocytic deficiency 18%

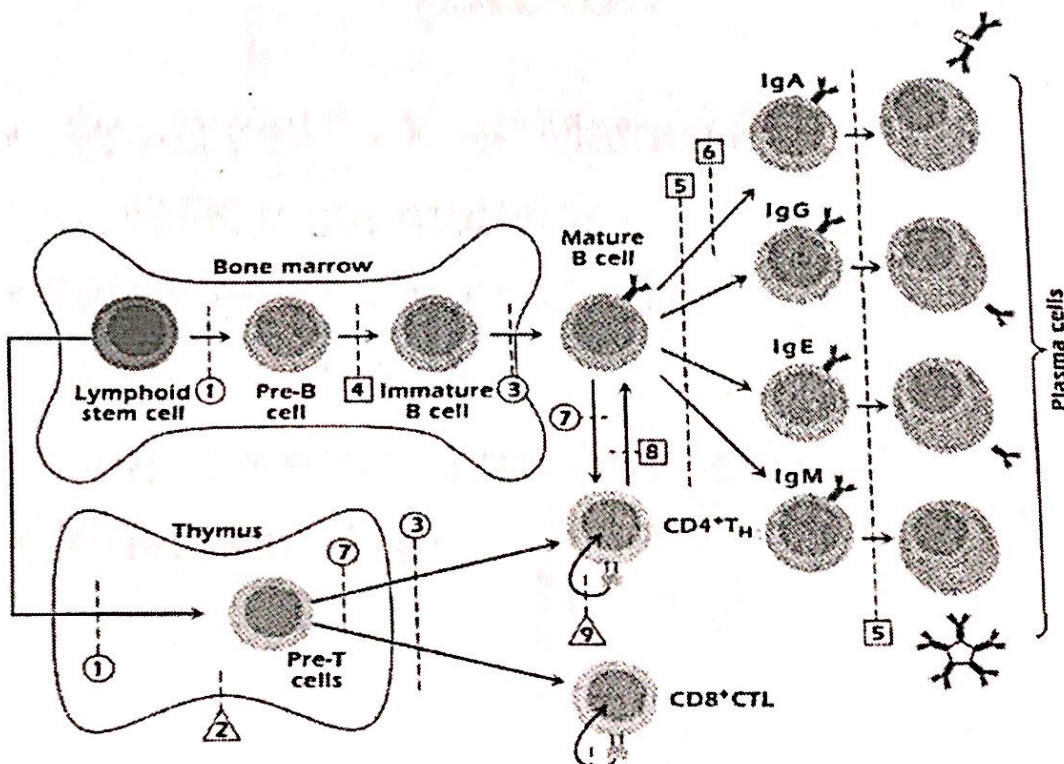
Complement deficiency 2%

→ 70% of ID has to do with B cells

All patients with recurrent infections should be screened for immunodeficiency

B cell bacterial

* All patients with recurrent viral infection should be screened for T cell immunodeficiency



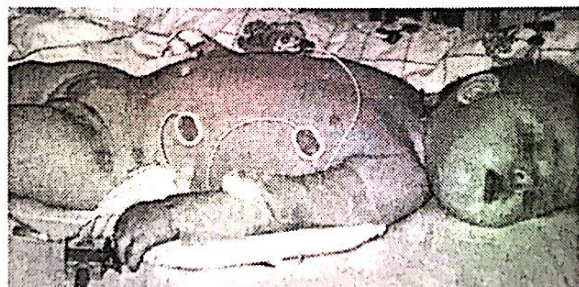
B cell ID

- Responsible for 50% of ID
- Recurrent bacterial otitis caused by *S. pneumonia* and *H. influenzae*
- Recurrent respiratory and gastrointestinal infection with pyogenic bac.

B cell ID linked diseases:

① Infantile Agammaglobulinemia (Bruton)

- Pre-B cell → X → Mature B
BTK gene mutation
- Absence of mature B cells
- Absence of all Ig
- Characteristic undeveloped tonsils
- Treatment: IVIg



② Common Variable ID diseases (CV)

- Mature B cell $\rightarrow$ X $\rightarrow$ plasma cell
- Absence of mature B cells
- Absence of all Ig
- Treated patients have normal life
- Treatment: IVIg.

* degraded deficiency: 10%, 40%, etc.

③ IgA deficiency



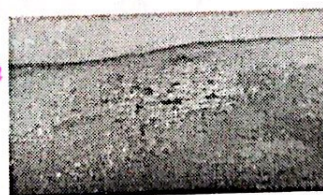
- Most common ID: 1 out of 800 in Europe
- Mature B $\rightarrow$ X $\rightarrow$ IgA
- Treated by antibiotics and not by IVIG because:

a. Commercial prep. Have low level of IgA (it's hard to isolate IgA)

b. Injected IgA does not reach the external surfaces

c. Allergy due to antiIgA

- Treated patients have normal life

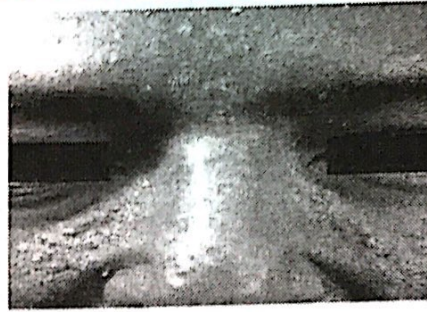


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HyperIgM syndrome

- Mutation in CD40L--- No B-T cell cooperation-----No class switch
- B & T have normal count
- Elevated IgM and low other classes

treatment: IVIg



T cell ID⁰

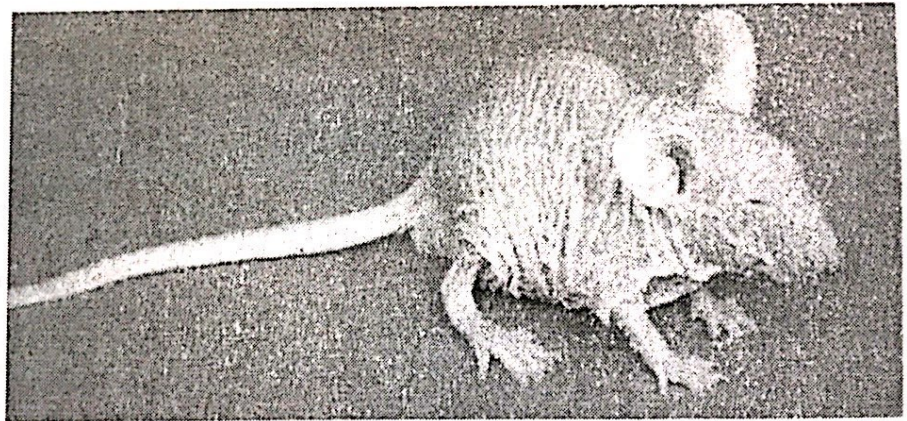
- Patients are susceptible to viral, fungal and protozoan infection and Opportunistic infection: Pneumocystis, CMV, candida, Mycobacterium, Toxoplasma
- Defect in T cells-----Defect in CD4+ cells-----
No class switch-----No IgG and only IgM to T-indep Ags.

Congenital thymic aplasia (DiGeorge syndrome)

- Failure to develop thymus, PTG, heart *linked also to parathyroid gland*
- Absence of mature T cells,
- Animal model: nude mouse
- B cells and serum Ig normal
- Patients fail to develop immune response to T-dep Ags but T-ind i.r is intact
- Hypocalcemia and heart defect
- Fetal thymus graft is not fully successful
- Should not be immunized with live attenuated vaccines

*treatment: parathyroid hormone

no calcium in blood
which affect bones



nu/nu gene (autosomal recessive)

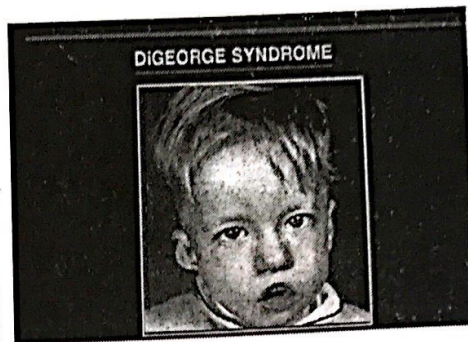
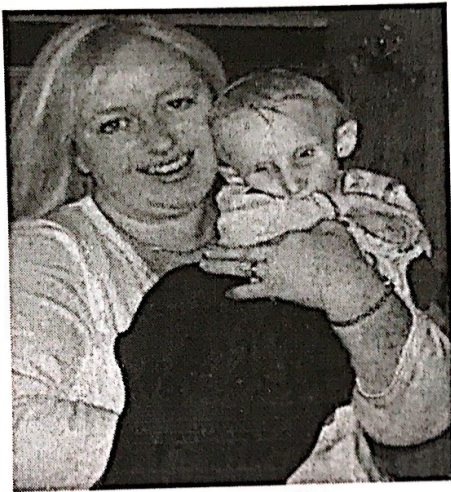
Hairless

Should be maintained in pathogen-free environment

T helper cell defect

Results in impaired cytotoxic T cell activity and Th-dependent B cell responses due to Th cell defect

Accept xenografts



② Autoimmune lymphoproliferative syndrome

- Massive increase in double negative immature CD4- and CD8-
- Massive proliferation of lymphoid cells (cancer)
- Patients have an increased number of double negative T cells
- Mutation of Fas or FasL ^{cause of disease}
- Most patients have one normal and one mutated Fas molecule
- Mouse model: lpr/lpr for fas and gld/gld for FasL deficiency
- Usually undetected until patient exposure to EBV ^{epstein-barr virus}-----
serious symptoms appear---often followed by Burkitts lymphoma

ID associated with both HI and CMI

- Sever combined ID or SCID
- Patients are susceptible to every type of infection
- Classified according to lym subsets in the blood

T-B+, T-B-, T+B-, T+B+

- The preferred treatment: T cell depleted BMT from matched sibling donor
- Patients Should not be immunized with live attenuated vaccines

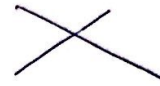
bone marrow transplantation

* normal count of B and T cells but no function



Fig 2-2.—Progressive varicella in infant with severe combined immunodeficiency.

T-B+



- Absence of T and NK cells but nonfunctional normal count of B cells
- Comprise 40-50% of SCID
- **x-linked mutation** in γ chain which is common to IL-2,4,7,9,15 receptors
- Knockout of IL-7 or R resemble SCID in animal model and not IL-2
- IL-2 knockout---some immune dysfunction----with normal B and T development and No SCID
- Autosomal recessive SCID
- Mutation in JAK Tyr kinase of signal transduction, IL-7

T-B-

enzyme

ADA Adenosine deaminase deficiency

- Def. lead to build up of toxic wastes causing symptoms (*no nucleotides*)
- The E is in purine salvage pathway
- Failure of both B and T cell develop---severe lymphopenia.
- *enzyme*
BMT is the best choice and E supplementation
- Patients lack a matched sibling bone marrow donor,
The first group to be treated with gene therapy

- 
- **PRP Purine nucleoside phosphorylase**
 - **in salvage pathway also**

Recombinase Def.

- **RAG1 and 2 deficiencies in the Pre-B and Pre-T cells but NK cells are normal**
- **Radiosensitivity because of loss of DNA repair**



T+B-

Omenn syndrome

- **Partial deficiency of RAG**
- **Fail to mount an IR to any Ag**
- **Th2----Cytokines----HyperIgE syndrome**
- **BMT success is low—Although patients are immunocompromised they should be treated with immunosuppressive therapy**

T+B+

- **Bare Lymphocyte Syndrome**
- Failure to express HLA class II due to def. in the gene for transcription factors
- Normal count of circulating B and T
- No HLA class II---No positive selection---
NO CD4+ cells-----No Ag presentation

Phagocytic Dysfunction

- **LAD II** Selectin L def. or missing on WBC
WBC cannot roll along EC
Defective in fucose metabolism
- **LAD I** missing B chain of integrin
Adhesion is impaired